

Technology Stack

Date	05 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	<i>HTML5, CSS3 (Custom Glassmorphism), FontAwesome</i>	<i>Chosen to build a lightweight, highly responsive user interface without framework overhead. Custom CSS variables enable a premium theme (champagne-yellow/cream), while FontAwesome provides structured icon integration.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python (Flask Framework)</i>	<i>Flask was chosen because it is a lightweight micro-framework with native support for Python libraries. It allows fast routing setups and direct integration with serialized machine learning models.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>SQLite & Flask-SQLAlchemy (ORM)</i>	<i>SQLite provides a zero-configuration, serverless relational database stored in a single file, making it highly portable. SQLAlchemy simplifies table mapping (User, Profile, Loan) and prevents SQL Injection.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Pinggy Tunneling</i>	<i>Chosen for instant, zero-setup public staging deployment. It tunnels the local port (5001) to a secure public HTTPS link (run.pinggy-free.link) so external clients and testers can run live predictions.</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Used to manage the source code repository, track file modifications, and coordinate pushes to the project's online profile (https://github.com/kvnvk9955-stack/Smart-Lender.git).</i>

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	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-Learn, XGBoost, SMOTE (imbalanced-learn), Seaborn</i>	<i>XGBoost was selected for its high classification accuracy. SMOTE is used to balance the target loan approval labels. Scikit-Learn handles scaling (StandardScaler), while Seaborn generates the dashboard plots.</i>